STAT 527 Advanced Regression Analysis
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Properties of Least Squares (Part I)

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Least Squares Estimator

We have seen that the least squares estimator is given by

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{y}.$$

Therefore, the fitted values are

$$\begin{aligned}\hat{\boldsymbol{y}} = \boldsymbol{X}\hat{\boldsymbol{\beta}} &= \boldsymbol{X}(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{y}, \\ &= P_X\boldsymbol{y}\end{aligned}$$

where

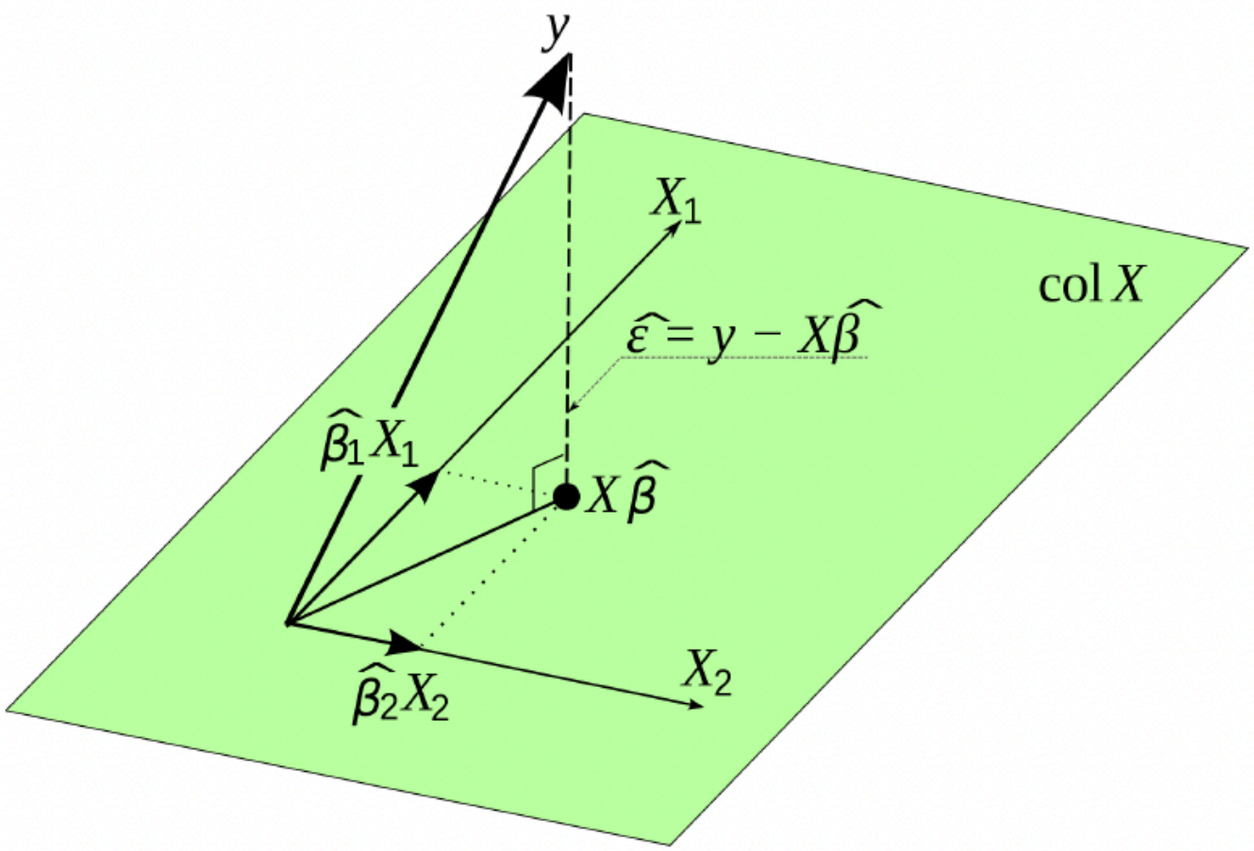
$$P_X = \boldsymbol{X}(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T$$

is the projection matrix onto the column space of $\boldsymbol{X}$. The residuals are given by

$$\hat{\boldsymbol{\epsilon}} = \boldsymbol{y} - \hat{\boldsymbol{y}} = \boldsymbol{y} - P_X\boldsymbol{y} = (I - P_X)\boldsymbol{y}.$$

Projection

Recall the geometric interpretation of least squares.



Projections

- Essentially, the least squares estimator divides the response vector $\boldsymbol{y}$ into two parts as follows:

$$\begin{aligned} \boldsymbol{y} &= \hat{\boldsymbol{y}} + (\boldsymbol{y} - \hat{\boldsymbol{y}}) \\ &= P_X \boldsymbol{y} + (I - P_X)\boldsymbol{y} \end{aligned}.$$

- The first part $P_X \boldsymbol{y}$ is the part of $\boldsymbol{y}$ that is fitted by the covariates $\boldsymbol{X}$.
- The second part is the residual part that is not explained by the covariates.

Review: Projection Matrices

- A square matrix P is a **projection** if it is symmetric and $P^2 = P$ (what if you drop the symmetric assumption? Consider rank-1 case).
- Equivalently, $P^\top (I - P) = 0$.
- By the spectral decomposition of symmetric matrices, there exist orthogonal Q such that $P = Q^\top \Lambda Q$, where

$$\Lambda = \begin{bmatrix} I & O \\ O & O \end{bmatrix}$$

  is a diagonal matrix.
- $X(X^\top X)^{-1} X^\top$ is a projection.

Orthogonality of Residuals

- We have already seen geometrically that residuals are orthogonal to fitted values, that is $\hat{\boldsymbol{y}}^T(\boldsymbol{y} - \hat{\boldsymbol{y}}) = 0$.
- This is also evident by noting $P_X^T(I - P_X) = 0$.
- Similarly, $\boldsymbol{X}^T(\boldsymbol{y} - \hat{\boldsymbol{y}}) = 0$ can be seen by noting that $\boldsymbol{X}^T(I - P_X) = 0$.

These imply that residuals are orthogonal to fitted values and orthogonal to the column space of X.

Covariance Matrix for a Random Vector

For a (random) vector

$$\boldsymbol{z} = \begin{pmatrix} z_1 \\ z_2 \\ \vdots \\ z_m \end{pmatrix}$$

the **mean vector** $\mathrm{E}(\boldsymbol{z})$ is just the vector of means, and the **variance-covariance matrix** or simply covariance matrix is

$$\mathrm{var}(\boldsymbol{z}) = \begin{pmatrix} \mathrm{var}(z_1) & \mathrm{cov}(z_1, z_2) & \cdots & \mathrm{cov}(z_1, z_m) \\ \mathrm{cov}(z_2, z_1) & \mathrm{var}(z_2) & & \mathrm{cov}(z_2, z_m) \\ \vdots & & \ddots & \vdots \\ \mathrm{cov}(z_m, z_1) & \mathrm{cov}(z_m, z_2) & \cdots & \mathrm{var}(z_m) \end{pmatrix}$$

Q: What are these for $\boldsymbol{\varepsilon}$? For $\boldsymbol{y}$?

Covariance Matrix for $\hat{\boldsymbol{\beta}}$

To find the variance-covariance matrix of $\hat{\boldsymbol{\beta}}$, we use the general result that for a constant matrix $\boldsymbol{A}$ and a random vector $\boldsymbol{z}$,

$$\mathrm{var}(\boldsymbol{A}\boldsymbol{z}) \;=\; \boldsymbol{A}\,\mathrm{var}(\boldsymbol{z})\,\boldsymbol{A}^T$$

Using this fact we obtain

$$\begin{aligned}\mathrm{var}(\hat{\boldsymbol{\beta}}) \;&=\; (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T(\sigma^2\boldsymbol{I})\boldsymbol{X}(\boldsymbol{X}^T\boldsymbol{X})^{-1} \\ &=\; \sigma^2(\boldsymbol{X}^T\boldsymbol{X})^{-1}\end{aligned}$$

[explain details ...]

Variance of Residuals

Consider the variance of the residuals $\hat{\boldsymbol{\varepsilon}} = \boldsymbol{y} - \hat{\boldsymbol{y}}$.

$$\begin{aligned}
\operatorname{var}(\hat{\boldsymbol{\varepsilon}}) &= var[\boldsymbol{y} - P_X \boldsymbol{y}] \\
&= var[(\boldsymbol{I} - P_X)\boldsymbol{y}] \\
&= \sigma^2 (\boldsymbol{I} - P_X)(I - P_X) = \sigma^2 (\boldsymbol{I} - P_X).
\end{aligned}$$

Estimating σ^2

The squared residuals are the squared deviations of the observations from their estimated expected values, which are useful in estimating the error variance $\sigma^2 = \operatorname{var}(\varepsilon_i)$.

Let RSS (Residual Sum of Squares) denote the sum of the squared least-squares residuals:

$$RSS = \sum_{i=1}^{n} \hat{\varepsilon}_i^2 = (\boldsymbol{y} - \boldsymbol{X}\hat{\beta})^\top (\boldsymbol{y} - \boldsymbol{X}\hat{\beta})$$

The following is an unbiased estimator of σ^2:

$$\hat{\sigma}^2 = \frac{RSS}{(n - p')}$$

Maximum Likelihood Estimators

If the errors are Gaussian, then we can estimate β and σ^2 using maximum likelihood estimation. The likelihood function is:

$$\ell(\boldsymbol{\beta}, \sigma^2) \propto \frac{1}{\sigma^n} \exp\left\{ -\frac{1}{2\sigma^2} (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})^\top (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}) \right\}.$$

By maximizing the log of the likelihood, we obtain:

$$\hat{\boldsymbol{\beta}}_{MLE} = \hat{\boldsymbol{\beta}}, \ \hat{\sigma}^2_{MLE} = \frac{RSS}{n}. \quad \textbf{Why?}$$

The MLE estimator for $\boldsymbol{\beta}$ it is the same as the least squares estimator and is unbiased but for σ^2 it is not unbiased.

Verification of MLE of the Variance

From the previous slide,

$$\log \ell(\beta, \sigma) = -\frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} RSS. \tag{1}$$

Take the derivative in σ^2 and set it to zero, we obtain

$$\sigma^2_{MLE} = \frac{1}{n} RSS. \tag{2}$$

Degrees of Freedom

The **residual degrees of freedom** is $n - p'$, and is defined by

$$\text{residual df} = \text{number of observations} - \text{number of regression coefficients in model}$$

Under the assumptions of the linear regression model, $E[\hat{\sigma}^2] = \sigma^2$.

Distributions of the Estimators

In addition to the standard assumptions, if the ε_i's are normally distributed and independent, then we can find the precise distributions of our estimators.

What is the distribution of $\hat{\boldsymbol{\beta}}$? What is the distribution of $\hat{\sigma}^2$?

Distributions of the Estimators

In addition to the standard assumptions, if the ε_i's are normally distributed and independent, then we can find the precise distributions of our estimators.

What is the distribution of $\hat{\boldsymbol{\beta}}$? What is the distribution of $\hat{\sigma}^2$?

$$\hat{\boldsymbol{\beta}} \sim N(\beta_0, \sigma^2(\boldsymbol{X}^T\boldsymbol{X})^{-1})$$

$$(n-p')\,\frac{\hat{\sigma}^2}{\sigma^2} \;\sim\; \chi^2(n-p').$$

Are they independent? Homework! ...Or see next slide

Verification of the Independence

$$P\epsilon \perp (I - P)\epsilon \tag{1}$$

$$\Rightarrow Py \perp (I - P)\epsilon \tag{2}$$

$$\Rightarrow \hat{y} \perp (I - P)\epsilon \tag{3}$$

$$\Rightarrow \hat{\beta} = (X^\top X)^{-1} X^\top \hat{y} \perp (I - P)\epsilon = \hat{\epsilon} \tag{4}$$

Variance Estimate

We can simply plug in this estimator of σ^2 to obtain a variance estimate of $\hat{\boldsymbol{\beta}}$.

Recall that the variance-covariance matrix of $\hat{\boldsymbol{\beta}}$ is given by

$$\text{var}(\hat{\boldsymbol{\beta}}) \;=\; \sigma^2(\boldsymbol{X}^T\boldsymbol{X})^{-1}$$

We can estimate this with

$$\widehat{\text{var}}(\hat{\boldsymbol{\beta}}) \;=\; \hat{\sigma}^2(\boldsymbol{X}^T\boldsymbol{X})^{-1}$$

Standard Error

The square root of an estimated variance is called a **standard error** (se). For example,

$$se(\hat{\beta}_1) = \sqrt{\widehat{\text{var}}(\hat{\beta}_1)} = \hat{\sigma} \sqrt{(\boldsymbol{X}^T \boldsymbol{X})^{-1}_{11}}$$

[Can find from variance-covariance matrix ...]

In R, $\hat{\sigma} = \sqrt{\hat{\sigma}^2}$ is called the *residual standard error*.

Thank you!

We gratefully acknowledge the contributions of previous course instructors and various online sources for some of the images and slides presented.